

1. CLO 2 (7 คะแนน) Fill in the reasons in the following proof sequence. Make sure you indicate which step(s) each derivation rule refers to.

Statements	Reasons
1. $p \wedge (q \vee r)$	given
2. $\neg(p \wedge q)$	given
3. $\neg p \vee \neg q$	De Morgan's Law (2)
4. $\neg q \vee \neg p$	Commutative (3)
5. $q \rightarrow \neg p$	Implication (4)
6. p	Simplification (1)
7. $\neg(\neg p)$	Double Negation (1)
8. $\neg q$	Modus Tollens (7,5)
9. $(q \vee r) \wedge p$	Commutative (1)
10. $q \vee r$	Simplification (9)
11. $r \vee q$	Commutative (10)
12. $\neg(\neg r) \vee q$	Double Negation (11)
13. $\neg r \rightarrow q$	Implication (12)
14. $\neg(\neg r)$	Modus Tollens (13,8)
15. r	Double Negation (14)
16. $p \wedge r$	Conjunction (15,6)

2. CLO 2 (5 คะแนน) Let x and y be numbers. Simplify the following statement using De Morgan's laws and double negation.

It is not the case that x is not greater than 3 and y is not found.

Ans

a is greater than 3 is $G(a)$

a is found $F(a)$

1 $\neg(G(x) \wedge \neg F(y))$ given

" x is greater than 3 or y is found."

2 $\neg(\neg G(x)) \vee \neg(\neg F(y))$ De Morgan's Law (1)

$G(x) \vee F(y)$ Double Negation (2)

3. CLO 2 (20 คะแนน) Recall Exercise 31 of Homework 1-1.

Suppose that all the following status reports are correct:

- Processor B is not working, and processor C is working.
- Processor A is working if and only if processor B is working.
- At least one of the two processors A and B is not working.

Let a = "A is working," b = "B is working," and c = "C is working."

- (3 คะแนน) Write each status report in terms of a , b , and c , using the symbols of formal logic.
- (3 คะแนน) Given the statements in part (A), what is the status of processor B? Justify your answer.
- (7 คะแนน) Given the statements in part (A), how would you justify the conclusion that processor C is working?
- (7 คะแนน) Given the statements in part (A), write a proof sequence to conclude that processor A is not working.

A Ans. stm. 1 $\neg b \wedge c$

stm. 2 $a \leftrightarrow b$

stm. 3 $\neg a \vee \neg b$

B. Let $\neg b \wedge c$ be true, we can deduce

if $\neg b \wedge c$ be true so c is true and b is false

that be: "Processor B is not Working"

D Ans. 1 $\neg b \wedge c$ given

2 $a \leftrightarrow b$ given

3 $\neg a \vee \neg b$ given

4 $\neg b$ Simplification (1)

5 $(a \rightarrow b) \wedge (b \rightarrow a)$ Biconditional Equivalence (2)

6 $a \rightarrow b$ Simplification (5)

7 $\neg a$ Modus Tollens (4, 6)

C Ans. from $\neg b \wedge c$, we can deduce that

b must be false and c must be true to give

that be true so: "Processor C is working"

4. CLO2(5 คะแนน) Write a proof sequence for the following assertion. Justify each step.

$$\left. \begin{array}{l} p \rightarrow \neg q \\ r \rightarrow (p \wedge q) \end{array} \right\} \Rightarrow \neg r$$

5. CLO2(5 คะแนน) Write a proof sequence for the following assertion. Justify each step.

$$\left. \begin{array}{l} p \rightarrow q \\ p \wedge r \end{array} \right\} \Rightarrow q \wedge r$$

6. CLO 2 (15 คะแนน) Write a proof sequence to establish that the assertion $p \Leftrightarrow p \vee p$ is a tautology.

7. CLO2(13 คะแนน) Write a proof sequence for the following assertion. Justify each step.

$$\neg(\neg p \rightarrow q) \vee (\neg p \wedge \neg q) \Rightarrow \neg p \wedge \neg q$$

8. CLO2(5 คะแนน) Write a proof sequence to show that the assertion $(a \rightarrow b) \wedge (a \wedge \neg b)$ is a contradiction.

9. CLO2(5 คะแนน) Is the assertion $a \rightarrow \neg a$ a contradiction? Write a proof sequence to justify your answer.

4. Ans

1. $p \rightarrow \neg q$ given
2. $r \rightarrow (p \wedge q)$ given
3. $\neg(r \wedge q) \rightarrow \neg p$ contraposition (1)
4. $q \rightarrow \neg p$ Double Negation (3)
5. $\neg(p \wedge q)$ De Morgan's Law (4)
6. $\neg r$ Modus Tollens (2, 5)

5. 1. $p \rightarrow q$ given

2. $p \wedge r$ given
3. p simplification (2)
4. q Modus ponens (1, 3)
5. r simplification (2)
6. $q \wedge r$ conjunction (4, 5)

6. Ans

1. $p \Leftrightarrow p \vee p$ given
2. $(p \rightarrow (p \vee p)) \wedge ((p \vee p) \rightarrow p)$ biconditional Equivalence (1)
3. $p \vee p \rightarrow p$ idempotent law
4. $(p \rightarrow p) \wedge (p \rightarrow p)$ substitution (3 $\rightarrow$ 2)
5. $T \wedge T$ Implication law
6. T conjunction

7. Ans

1. $\neg(p \rightarrow \neg q) \vee (\neg p \wedge \neg q)$ given
2. $\neg(\neg p) \vee \neg q \vee (\neg p \wedge \neg q)$ implication law (1)
3. $(p \vee \neg p) \vee (\neg p \wedge \neg q)$ double negation (2)
4. $p \vee (\neg q \vee (\neg p \wedge \neg q))$ association law (3)
5. $p \vee ((\neg q \vee \neg p) \wedge (\neg q \vee \neg q))$ distribution law (4)
6. $p \vee ((\neg q \vee \neg p) \wedge \neg q)$ idempotent law (5)
7. $(p \vee (\neg q \vee \neg p)) \wedge (p \vee \neg q)$ distribution law (6)
8. $((p \vee \neg p) \vee \neg q) \wedge (p \vee \neg q)$ commutative (7)
9. $(T \vee \neg q) \wedge (p \vee \neg q)$ complement law (8)
10. $T \wedge (p \vee \neg q)$ domination law (9)
11. $p \vee \neg q$ identity law (10)
12. $\neg q \vee p$ commutative law (11)
13. $q \Rightarrow p$ implication law (12)

8. Ans

1. $(a \rightarrow b) \wedge (a \wedge \neg b)$ given
2. $(\neg a \vee b) \wedge (a \wedge \neg b)$ implication law (1)
3. $(\neg a \vee b) \wedge a \wedge \neg b$ associative law (2)
4. $(\neg a \wedge a) \vee (b \wedge a \wedge \neg b)$ distribution law
5. $(\neg a \wedge a) \vee (b \wedge \neg b)$ commutative law (4)
6. $(F \wedge \neg b) \vee (F \wedge a)$ complement law (5)
7. $F \vee F$ domination law (6)
8. F identity law (7)

9. Ans

1. $a \rightarrow \neg a$ given
2. $\neg a \vee \neg a$ implication law (1)
3. $\neg a$ idempotent law (2)

10. CLO 1 (20 คะแนน) แสดงว่า argument ที่กำหนดให้มีความสมเหตุสมผล สอดคล้องกัน นั่นคือ แสดงว่า จากสมมติฐานที่กำหนดให้ สามารถอนุมานได้เป็นบทสรุป (conclusion) ที่กำหนดให้ได้

- All computer science majors must take CS101.
- CS101 students study discrete structures.
- So, all computer science majors must study discrete structures.

กำหนด predicate และ โดเมนของ x ให้ใช้งานดังนี้

- $CS(x)$: " x is CS major."
- $CS101(x)$: " x takes CS101".
- $S(x)$: " x studies discrete structures".

(Domain of x is set of all TU students.)

- Ans
1. $\forall x [CS(x) \rightarrow CS101(x)]$ given
 2. $\forall x [CS101(x) \rightarrow S(x)]$ given
 3. $CS(a) \rightarrow CS101(a)$ Universal instantiation (1)
 4. $CS101(a) \rightarrow S(a)$ Universal instantiation (2)
 5. $CS(a) \rightarrow S(a)$ Hypothetical syllogism (1, 4)
 6. $\forall x [S(x) \rightarrow S(x)]$ Universal generalization (5)

ทุก proof sequence แสดงให้เห็นว่า สมเหตุสมผล